

## Experiment 7

## End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

|                     |                                               |             |
|---------------------|-----------------------------------------------|-------------|
| Degree & Branch     | B.Tech Artificial Intelligence & Data Science | Semester V  |
| Subject Code & Name | CS3807 – Deep Learning Laboratory             | AY: 2026–27 |

## 1 Objective

The objective of this experiment is to build an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modelling. Four models were implemented on the MNIST handwritten digit dataset and evaluated under a common protocol:

1. a fully connected autoencoder (FC-AE);
2. a convolutional autoencoder (CAE);
3. a denoising convolutional autoencoder trained with Gaussian and salt-and-pepper corruption; and
4. a variational autoencoder (VAE) with a two-dimensional latent space.

Reconstruction quality is compared using MSE, MAE and SSIM, and the VAE latent space is analysed through visualisation, random generation and interpolation.

## 2 Dataset and Experimental Setup

**Dataset.** MNIST contains grayscale images of handwritten digits 0–9 of size  $28 \times 28 \times 1$  [4]. Pixel values were scaled from  $[0, 255]$  to  $[0, 1]$ . The 60,000 training images were divided into 54,000 training and 6,000 validation images, and all reported metrics are computed on the 10,000 test images. The full dataset was used rather than the 10,000/2,000 laboratory subset.

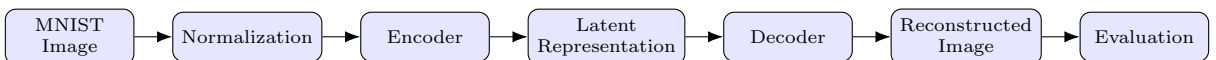
**Inputs and targets.** For the FC-AE each image is flattened to  $x \in \mathbb{R}^{784}$ ; for the convolutional models the spatial structure  $x \in \mathbb{R}^{28 \times 28 \times 1}$  is retained. The original image is the reconstruction target; digit labels are used only for visualisation and analysis, never for training.

**Common training settings.** All models were implemented in Keras/TensorFlow [6, 7] and use the Adam optimiser with a learning rate of  $10^{-3}$  and a batch size of 128.

**Table 1:** Models studied in this experiment.

| Model              | Latent representation                             | Parameters | Training time (s)      |
|--------------------|---------------------------------------------------|------------|------------------------|
| Fully Connected AE | 16-D vector                                       | 211,040    | 48.67                  |
| Convolutional AE   | $7 \times 7 \times 64$ feature map (3,136 values) | 74,497     | 78.91                  |
| Denoising CAE      | $7 \times 7 \times 64$ feature map (3,136 values) | 74,497     | 51.21 ( $\sigma=0.2$ ) |
| VAE                | 2-D Gaussian ( $\mu, \log \sigma^2$ )             | 134,165    | 102.03                 |

The overall pipeline followed for every model is:



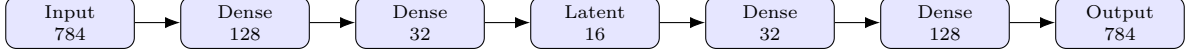
### 3 Methodology

#### 3.1 Autoencoder principle

An autoencoder [1] maps the input to a lower-dimensional code with an encoder,  $z = f_\theta(x)$ , and reconstructs the input with a decoder,  $\hat{x} = g_\phi(z)$ . The FC-AE, CAE and denoising CAE were trained with binary cross-entropy (BCE) between  $x$  and  $\hat{x}$ ; their quality is reported with the metrics of Section 3.6.

#### 3.2 Fully connected autoencoder

The image is flattened to 784 values and passed through the architecture below, with ReLU in the hidden layers and a sigmoid output. It was trained for 20 epochs with BCE loss.



#### 3.3 Convolutional autoencoder

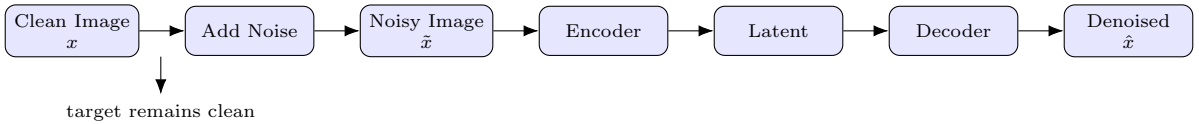
The CAE keeps the spatial layout of the image, using convolution and max-pooling in the encoder and up-sampling in the decoder. It was trained for 20 epochs with BCE loss.

```

Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')          -> 7x7x64 latent feature map
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
  
```

#### 3.4 Denoising autoencoder

A denoising autoencoder [3] receives a corrupted image  $\tilde{x}$  but is trained to reconstruct the clean image  $x$ , so that the clean image remains the target.



Two corruption types were used with the same architecture as the CAE:

- **Gaussian noise:**  $\tilde{x} = \text{clip}(x + n, 0, 1)$  with  $n \sim \mathcal{N}(0, \sigma^2)$  and  $\sigma \in \{0.1, 0.2, 0.3\}$ . A separate denoising CAE was trained for each  $\sigma$ ; the  $\sigma = 0.2$  model is used for the qualitative figure.
- **Salt-and-pepper noise:** a proportion  $p = 0.1$  of the pixels is replaced by 0 or 1.

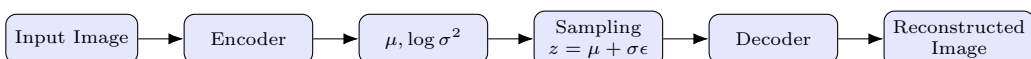
#### 3.5 Variational autoencoder

The VAE [2] encoder outputs the parameters of a Gaussian,  $q_\phi(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x)))$ , and a latent vector is drawn with the reparameterization trick

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

The loss combines a BCE reconstruction term with the analytic KL divergence to the prior  $p(z) = \mathcal{N}(0, I)$ :

$$\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}, \quad \mathcal{L}_{KL} = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$



The latent dimension is  $z \in \mathbb{R}^2$  so that the latent space can be plotted directly. The architecture is:

- **Encoder:** Conv2D(32, stride 2)  $\rightarrow$  Conv2D(64, stride 2)  $\rightarrow$  Flatten  $\rightarrow$  Dense(16)  $\rightarrow [\mu, \log \sigma^2] \rightarrow$  Sampling.
- **Decoder:** Dense( $7 \cdot 7 \cdot 64$ )  $\rightarrow$  Reshape  $\rightarrow$  Conv2DTranspose(64, stride 2)  $\rightarrow$  Conv2DTranspose(32, stride 2)  $\rightarrow$  Conv2DTranspose(1, sigmoid).

The VAE was trained for 30 epochs.

### 3.6 Evaluation metrics

Metrics are computed on the test set, with  $N$  images of  $D$  pixels each:

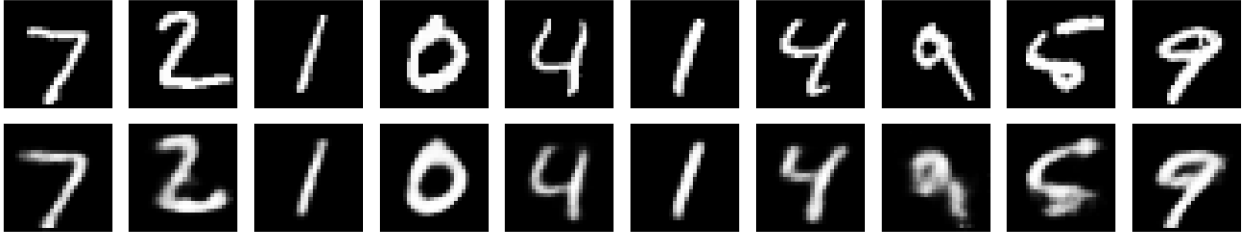
$$MSE = \frac{1}{ND} \sum_{i=1}^N \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2, \quad MAE = \frac{1}{ND} \sum_{i=1}^N \sum_{j=1}^D |x_{ij} - \hat{x}_{ij}|.$$

The structural similarity index (SSIM) [5] compares local luminance, contrast and structure between the original and reconstructed image; it is computed per image and averaged over the test set. Lower MSE/MAE and higher SSIM (maximum 1) indicate better reconstruction. For the denoising models, the denoised output is compared with the clean image.

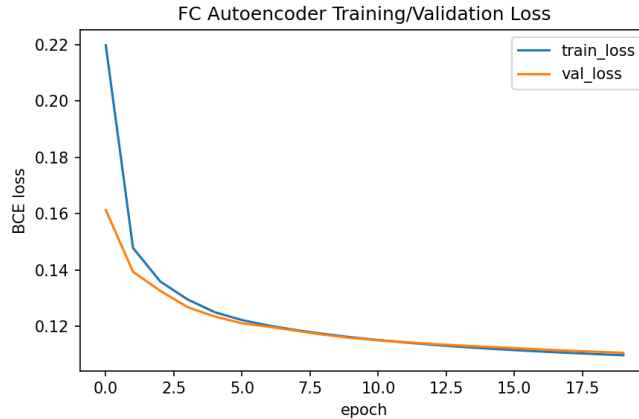
## 4 Results and Analysis

### 4.1 Fully connected autoencoder

Figure 1 shows ten test digits (top row) and their FC-AE reconstructions (bottom row); Figure 2 shows the training and validation BCE loss.



**Figure 1:** FC-AE: original test images (top) and reconstructions (bottom).



**Figure 2:** FC-AE training and validation BCE loss.

**Table 2:** FC-AE test-set reconstruction metrics (latent dimension 16).

| Metric    | Value    |
|-----------|----------|
| Test MSE  | 0.015567 |
| Test MAE  | 0.045359 |
| Mean SSIM | 0.8245   |

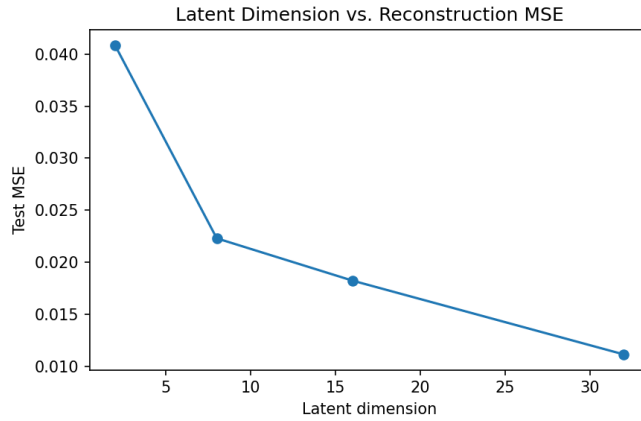
**Inference.** The reconstructions preserve the overall shape of most digits. Simple, high-contrast strokes (the 1s, the 0 and the 7) are reproduced well, while digits with curved or thin structure lose detail: the 2 and the two 4s are visibly blurred, and the thin-tailed 9 and the 5 are distorted into smudged shapes. A 16-value code must summarise the whole image, and a decoder trained with a pixel-wise loss tends to average over plausible strokes, which produces blur. The loss curves fall from about 0.22 (training) and 0.16 (validation) in the first epoch to about 0.11 for both, and the two curves stay almost on top of each other. The model therefore converges without evidence of overfitting; the validation loss lies below the training loss in the first epochs because the training loss is averaged over an epoch in which the model is still improving quickly. The curves are still decreasing slowly at epoch 20, so longer training could give a small further gain.

## 4.2 Effect of latent dimension

The FC-AE was retrained with  $d_z \in \{2, 8, 16, 32\}$ . Only the layers adjacent to the bottleneck change, so the parameter count stays almost constant.

**Table 3:** Effect of latent dimension on FC-AE reconstruction (test set).

| Latent dimension | Compression ( $784/d_z$ ) | MSE      | SSIM   | Parameters |
|------------------|---------------------------|----------|--------|------------|
| 2                | $392\times$               | 0.040862 | 0.5403 | 210,130    |
| 8                | $98\times$                | 0.022291 | 0.7509 | 210,520    |
| 16               | $49\times$                | 0.018226 | 0.7939 | 211,040    |
| 32               | $24.5\times$              | 0.011130 | 0.8754 | 212,080    |

**Figure 3:** Latent dimension versus test reconstruction MSE.

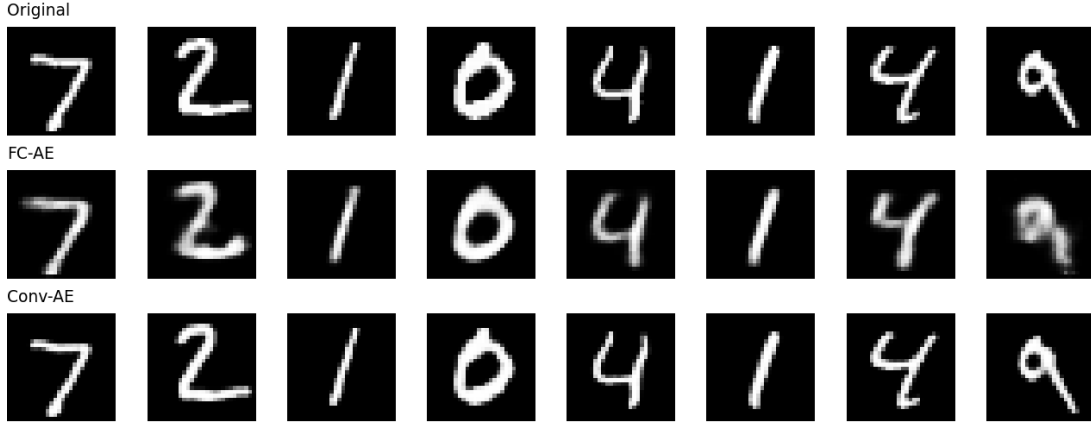
**Inference.** Increasing the latent dimension from 2 to 32 lowers the test MSE by about 73% (0.0409 to 0.0111) and raises SSIM from 0.540 to 0.875. The largest gain comes from the first increase (2 to 8); beyond that, each added dimension buys less, which gives the flattening curve in Figure 3. Because the parameter count is essentially unchanged, the improvement comes from the larger information capacity of the code, not from a larger model. This is the compression–fidelity trade-off: a very small code gives strong compression ( $392\times$  at  $d_z = 2$ ) but blurry, poorly preserved digits, while a larger code reconstructs better at the cost of compression. *Note:* the  $d_z = 16$  entry (MSE 0.0182, SSIM 0.794) comes from this separate study run and differs slightly from the main FC-AE run in Table 2 (MSE 0.0156, SSIM 0.825), so differences of that size between neighbouring settings should be read with caution.

### 4.3 Fully connected versus convolutional autoencoder

Both models were trained and tested on the same images.

**Table 4:** FC-AE versus CAE on the test set.

| Model              | MSE      | MAE      | SSIM   | Parameters |
|--------------------|----------|----------|--------|------------|
| Fully Connected AE | 0.015567 | 0.045359 | 0.8245 | 211,040    |
| Convolutional AE   | 0.001154 | 0.010152 | 0.9880 | 74,497     |

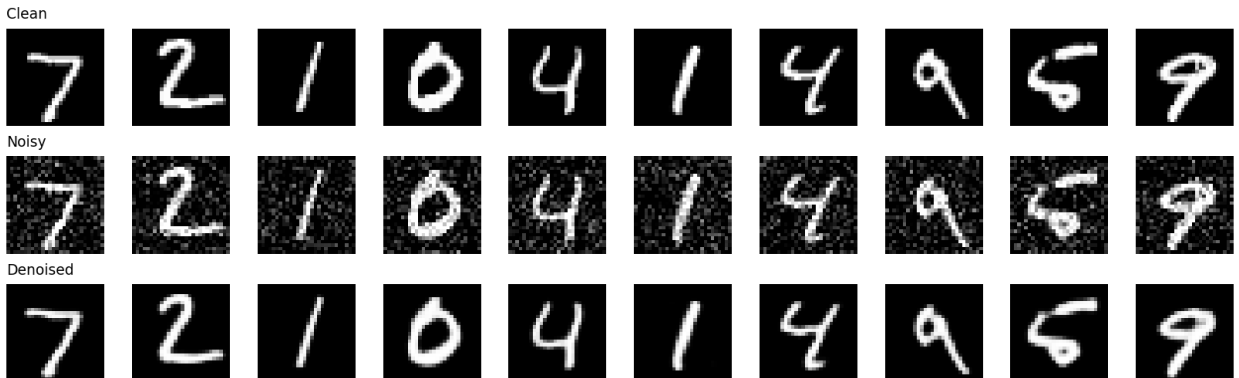


**Figure 4:** Original images (top), FC-AE reconstructions (middle) and CAE reconstructions (bottom).

**Inference.** The CAE reduces MSE by about  $13.5\times$  and MAE by about  $4.5\times$ , and raises SSIM from 0.825 to 0.988, while using about  $2.8\times$  fewer parameters. Visually (Figure 4) the CAE keeps stroke thickness, sharp edges and the thin tail of the final 9, which the FC-AE smears into a blur. Convolution exploits the spatial locality of images: small filters with shared weights detect local stroke patterns wherever they occur, whereas the first FC layer treats all 784 pixels as unrelated inputs and needs a separate weight for each. The CAE trains about  $1.6\times$  longer per run (78.9 s versus 48.7 s) because convolutions are more expensive per parameter. *Caveat:* the two models do not have equal bottlenecks. The CAE latent map has  $7 \times 7 \times 64 = 3,136$  values, four times the input size, whereas the FC-AE code has 16. Part of the gap therefore reflects the wider code and not convolution alone; even the 32-D FC-AE in Table 3 has about  $10\times$  higher MSE than the CAE.

### 4.4 Denoising autoencoder

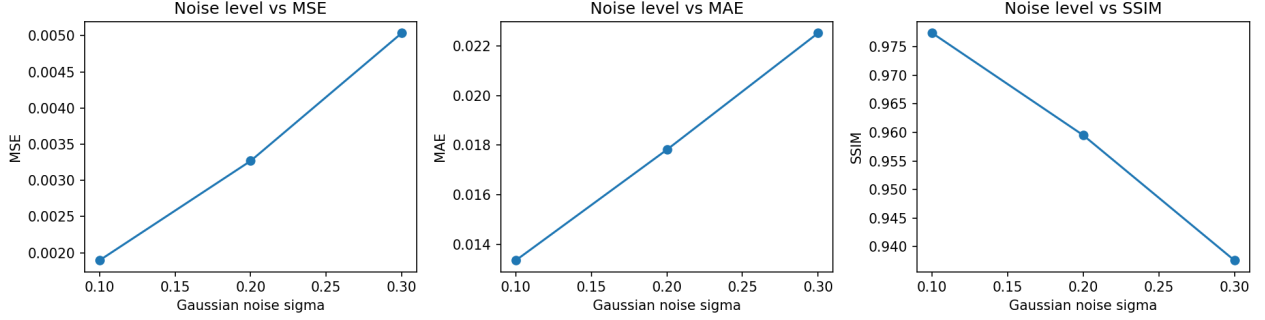
**Gaussian noise.** Figure 5 shows clean, noisy ( $\sigma = 0.2$ ) and denoised images.



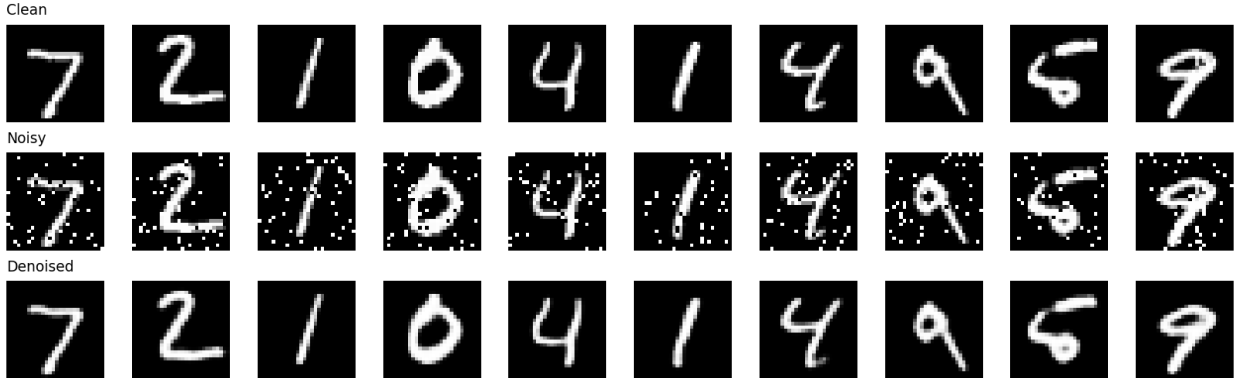
**Figure 5:** Gaussian denoising ( $\sigma = 0.2$ ): clean (top), noisy (middle) and denoised (bottom) test images.

**Table 5:** Denoising CAE performance for each Gaussian noise level (test set, denoised output versus clean image).

| Noise level $\sigma$ | MSE      | MAE      | SSIM   |
|----------------------|----------|----------|--------|
| 0.1                  | 0.001895 | 0.013360 | 0.9774 |
| 0.2                  | 0.003269 | 0.017817 | 0.9595 |
| 0.3                  | 0.005036 | 0.022511 | 0.9376 |

**Figure 6:** Gaussian noise level versus MSE, MAE and SSIM (separate axes for each metric).

**Salt-and-pepper noise.** A denoising CAE was also trained with salt-and-pepper corruption at  $p = 0.1$  (Figure 7), giving test MSE 0.002999, MAE 0.015978 and SSIM 0.9682.

**Figure 7:** Salt-and-pepper denoising ( $p = 0.1$ ): clean (top), noisy (middle) and denoised (bottom) test images.

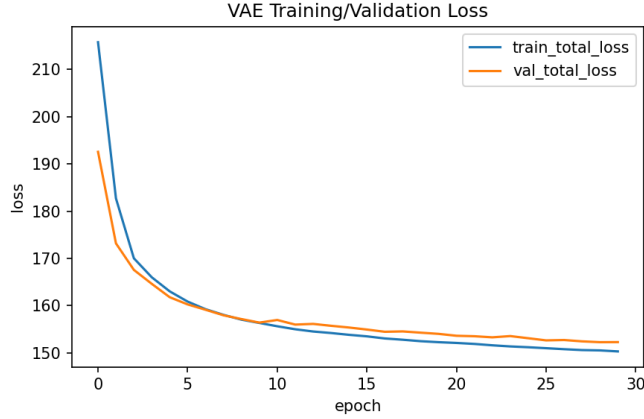
**Inference.** In the noisy inputs the background is speckled and strokes are partly broken, yet the denoised outputs show a clean black background and continuous strokes, and every digit remains recognisable and matches its clean original; only slight softening of stroke edges remains. As  $\sigma$  grows from 0.1 to 0.3, MSE rises about  $2.7\times$  (0.0019 to 0.0050), MAE rises from 0.0134 to 0.0225 and SSIM falls from 0.977 to 0.938 (Figure 6). The degradation is steady and gradual, with no collapse. Heavier noise hides more of the stroke information, so the network has to infer more of the digit from context and the residual error grows. Even at  $\sigma = 0.3$  the denoising CAE (MSE 0.0050) is better than the clean-input FC-AE (0.0156), and even light noise costs accuracy relative to the noise-free CAE (MSE 0.001154 versus 0.001895 at  $\sigma = 0.1$ , about 64% higher). The salt-and-pepper result at  $p = 0.1$  (SSIM 0.968) falls between the  $\sigma = 0.1$  and  $\sigma = 0.2$  Gaussian models, although the two corruption types are not directly comparable; only one salt-and-pepper level was evaluated. Each model was tested at the noise level it was trained on. Overall, the denoising autoencoder recovers the major structure of the digits.

#### 4.5 Variational autoencoder

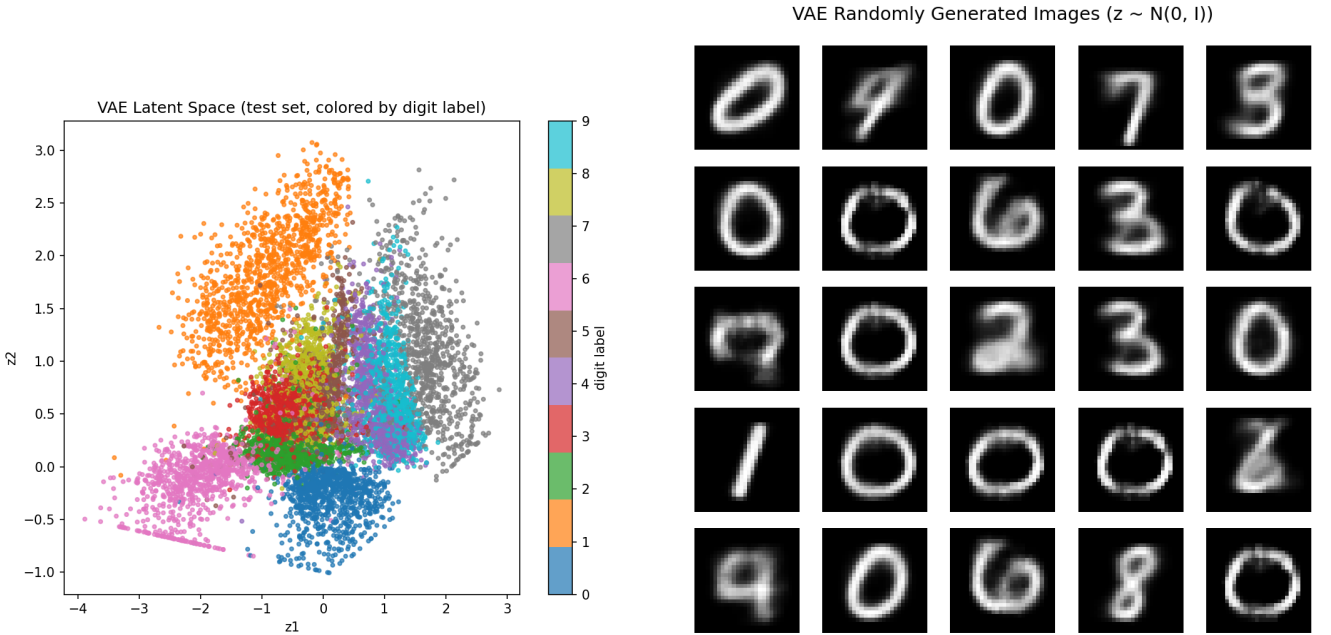
Table 6 lists the VAE results. The loss values are per image, with the reconstruction term summed over pixels, at the final training epoch.

**Table 6:** VAE loss components and test-set metrics.

| VAE metric                | Value    |
|---------------------------|----------|
| Reconstruction loss (BCE) | 143.7238 |
| KL loss                   | 6.5676   |
| Total loss                | 150.2913 |
| Test MSE                  | 0.041170 |
| Test MAE                  | 0.097289 |
| Mean SSIM                 | 0.5446   |

**Figure 8:** VAE training and validation total loss (reconstruction + KL).

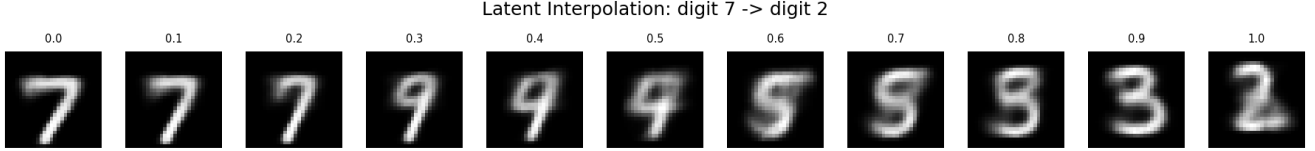
**Inference (training).** Both curves fall steeply in the first few epochs (from about 216 training / 193 validation) and flatten near 150 (training) and 152 (validation) by epoch 30. The small, stable gap of about 2 units (about 1.3%) shows only mild overfitting. The reconstruction term accounts for about 96% of the total loss and the KL term for about 4%. Figure 8 shows the total loss; the two terms are reported separately in Table 6. The VAE’s reconstruction metrics (MSE 0.0412, SSIM 0.545) are much worse than those of the CAE, mainly because the code has only two dimensions; this is examined further in Section 5.2.

**(a)** Encoded test images coloured by digit label.**(b)** Images generated from  $z \sim \mathcal{N}(0, I)$ .**Figure 9:** VAE latent space and random generation.

**Inference (latent space).** Even though labels were never used in training, digits form recognisable regions in

the  $(z_1, z_2)$  plane (Figure 9a). The 0s (bottom centre), 1s (top, a broad region), 6s (lower left) and 7s (right) are largely separate. In contrast, 4s and 9s overlap heavily, and 2, 3, 5 and 8 mix in the centre. Visually similar digits are placed close together because the VAE groups images by pixel similarity, and two dimensions are not enough to separate all ten classes. The codes span roughly  $z_1 \in [-4, 3]$  and  $z_2 \in [-1, 3]$ , so they occupy the region covered by the standard-normal prior, although the cloud is not perfectly centred (it is shifted towards positive  $z_2$ ). The small KL term (6.57 nats in total) is consistent with a posterior that is moderately close to the prior.

**Inference (generation).** Sampling  $z \sim \mathcal{N}(0, I)$  and decoding gives recognisable digits (0, 1, 3, 6, 7, 8 and some 4/9-like shapes; Figure 9b) with visible variation in slant and thickness. However, quality and diversity are limited: about half of the 25 samples (roughly 12) are 0s, several samples are blurry or ambiguous hybrids, and the strokes are soft compared with real digits. Prior samples fall most often in the large central region that decodes to rings, and regions between clusters decode to hybrids of neighbouring digits. The low-dimensional code and the likelihood-based loss also give blurred outputs. The VAE can therefore generate new plausible digits, but not with high fidelity or an even class balance.

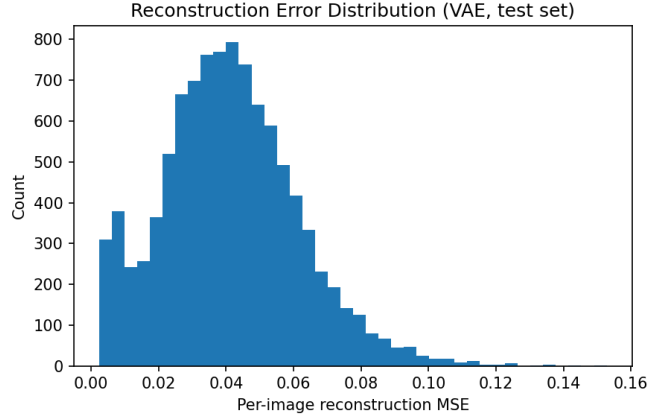


**Figure 10:** Latent interpolation between a test 7 ( $\alpha = 0$ ) and a test 2 ( $\alpha = 1$ ).

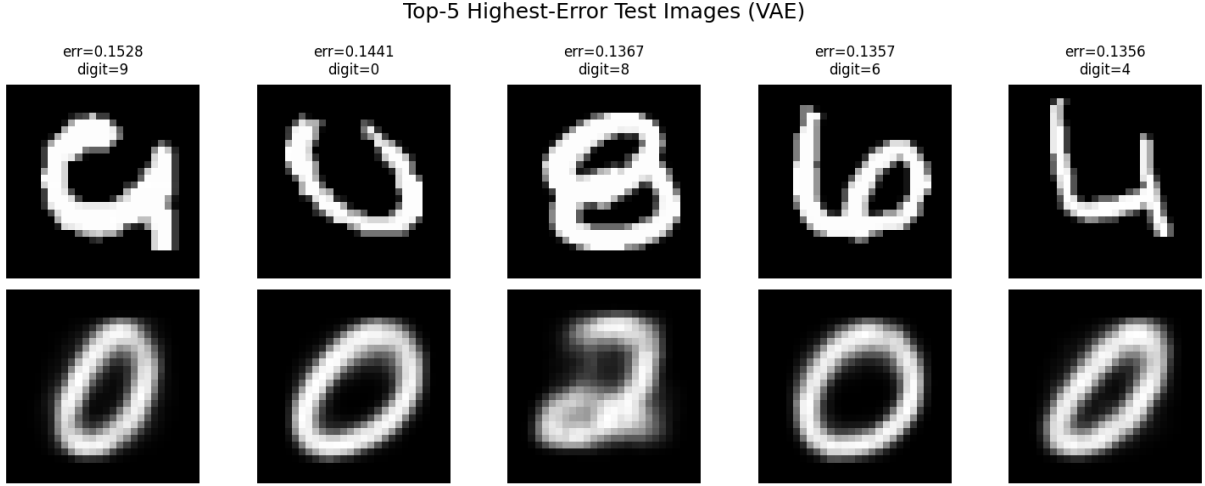
**Inference (interpolation).** The decoded image changes gradually as  $\alpha$  goes from 0 to 1 (Figure 10), with no abrupt jumps or noise. However, the path does not morph a 7 directly into a 2: it passes through other digits, roughly  $7 \rightarrow 9/4 \rightarrow 5 \rightarrow 3 \rightarrow 2$ . This agrees with the latent map, where a straight line from the 7 region to the 2 region crosses the 4/9, 5 and 3 regions. The latent space is thus continuous, since the decoder produces digit-like images along the whole path, but it is only weakly organised by shape, as expected from a 2-D code.

#### 4.6 Reconstruction-error analysis (VAE)

The per-image error is  $e_i = \frac{1}{D} \sum_j (x_{ij} - \hat{x}_{ij})^2$ . Figure 11 shows its distribution over the test set: mean 0.0412, standard deviation 0.0202, minimum 0.0024 and maximum 0.1528.



**Figure 11:** Distribution of per-image reconstruction MSE (VAE, test set).



**Figure 12:** The five test images with the highest VAE reconstruction error (top: original, bottom: reconstruction).

**Table 7:** The five highest-error test samples (VAE).

| Rank | Test index | Digit label | MSE      |
|------|------------|-------------|----------|
| 1    | 2293       | 9           | 0.152791 |
| 2    | 3818       | 0           | 0.144063 |
| 3    | 6161       | 8           | 0.136716 |
| 4    | 3835       | 6           | 0.135662 |
| 5    | 4269       | 4           | 0.135649 |

**Inference.** Most images have an error between about 0.02 and 0.06 (mean  $\pm$  one standard deviation is 0.021–0.061), with a peak near 0.04 and a right-skewed tail. A small secondary mode appears at very low error (0.005–0.01); this probably corresponds to simple, thin digits such as 1, but the per-class error was not computed. The worst cases (up to 0.153) are about 3.7 times the mean, so they are rare outliers. Figure 12 shows that these are unusual or malformed handwriting: a hooked, G-like 9, a 0 written as an open U, a heavy double-loop 8, a narrow tilted 6 and an open, skewed 4. Four of the five are reconstructed as a generic tilted oval resembling a 0 and the 8 as a blocky blob, since a two-dimensional code cannot represent atypical shapes and the decoder falls back to a typical form. Thick strokes and misaligned pixels also raise the pixel-wise error. The relationship between error and visual quality is clear at the extremes, where high MSE means an unrecognisable reconstruction. But MSE is not a perfect proxy for semantic correctness: the open-U 0 is reconstructed as a reasonable 0 yet still has a large error, because a smooth oval does not overlap the original strokes.

## 5 Consolidated Results

**Table 8:** Consolidated test-set results for all models. The denoising CAE row uses  $\sigma = 0.2$  and compares the denoised output with the clean image.

| Model                            | MSE      | MAE      | SSIM   | Parameters | Time (s) |
|----------------------------------|----------|----------|--------|------------|----------|
| FC Autoencoder                   | 0.015567 | 0.045359 | 0.8245 | 211,040    | 48.67    |
| Conv. Autoencoder                | 0.001154 | 0.010152 | 0.9880 | 74,497     | 78.91    |
| Denoising CAE ( $\sigma = 0.2$ ) | 0.003269 | 0.017817 | 0.9595 | 74,497     | 51.21    |
| VAE                              | 0.041170 | 0.097289 | 0.5446 | 134,165    | 102.03   |

### 5.1 Error and visual quality across models

The ranking by MSE (CAE < denoising CAE < FC-AE < VAE) is the same as the ranking by MAE and by SSIM, and it matches the visual quality: the CAE outputs are nearly indistinguishable from the originals, the FC-AE and the VAE outputs are blurred, and the VAE outputs at the tail are unrecognisable. The denoising CAE is worse than

the noise-free CAE because it solves a harder task (recovering the digit from a corrupted input), yet it is still better than the FC-AE reconstructing clean images.

## 5.2 Deterministic versus probabilistic latent representations

A conventional autoencoder maps each image to a single point, and nothing constrains where codes lie, so there is no distribution to sample from. The VAE maps each image to a Gaussian and its KL term pulls these distributions towards  $\mathcal{N}(0, I)$ , which yields a compact, continuous latent space that can be sampled (Figure 9b) and interpolated (Figure 10). The VAE’s poor reconstruction is mostly a capacity effect and not a cost of the probabilistic formulation: the FC-AE with  $d_z = 2$  obtains almost the same scores (MSE 0.0409, SSIM 0.540) as the VAE (MSE 0.0412, SSIM 0.545). This comparison is only indicative, since the architectures differ (fully connected versus convolutional). The trade-off is therefore that deterministic autoencoders with larger codes give the best reconstruction, while the VAE gives generative ability with a regularised latent space at lower reconstruction fidelity for the same code size.

## 6 Conclusion

- The FC-AE (16-D code) converged without overfitting (test MSE 0.0156, SSIM 0.825) but produced blurred reconstructions of thin or curved digits.
- Reconstruction quality improves with latent dimension, with strongest gains between 2 and 8 dimensions (MSE 0.0409 to 0.0223) and diminishing returns per added dimension.
- The CAE was far more accurate (MSE 0.00115, SSIM 0.988) with  $2.8\times$  fewer parameters, helped by exploiting spatial locality and by its wider  $7 \times 7 \times 64$  latent map.
- Denoising CAEs recovered the digit structure under both Gaussian and salt-and-pepper noise; quality degraded gradually as  $\sigma$  increased from 0.1 to 0.3 (SSIM 0.977 to 0.938).
- The 2-D VAE learned a partially class-organised latent space that supports sampling and smooth interpolation, but its generated and reconstructed images are blurry (SSIM 0.545), and unusual handwriting gives the largest errors.

## References

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